

Eigen shit

Eigenvalues + Eigenvectors

An eigenvector $\underline{x}$ of an $n \times n$ matrix A is one such that $A\underline{x} = \lambda \underline{x}$, where λ is a scalar, called an eigenvalue. The zero vector cannot be an eigenvector.

If we treat A as a linear transformation, this transformation scales $\underline{x}$ by λ , without changing its direction.

Finding Eigenvalues:

Use characteristic polynomial $\det(A - \lambda I)$ and set it equal to zero, and solve for λ

2x2 example (ugly)

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

$$\det(A - \lambda I) = \det \begin{pmatrix} 1-\lambda & 2 \\ 3 & 4-\lambda \end{pmatrix} = 0$$

$$(1-\lambda)(4-\lambda) - 6 = 0$$

$$\lambda^2 - 5\lambda - 2 = 0$$

$$\lambda = \frac{5 \pm \sqrt{25+8}}{2} = \frac{5}{2} \pm \frac{\sqrt{33}}{2}$$

$$\therefore \text{Eigenvalues } \lambda_1 = \frac{5+\sqrt{33}}{2}, \lambda_2 = \frac{5-\sqrt{33}}{2}$$



Neither eigenvalue is a repeated root of the characteristic polynomial.

This means that both λ_1 and λ_2 have an algebraic multiplicity of 1.

3x3 example

$$A = \begin{pmatrix} 3 & 1 & 3 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\det(A - \lambda I) = \det \begin{pmatrix} 3-\lambda & 1 & 3 \\ 0 & 3-\lambda & 0 \\ 0 & 0 & 1-\lambda \end{pmatrix} = 0$$

$$(1-\lambda)(3-\lambda)(3-\lambda) = 0$$

$$\therefore \text{Eigenvalues } \lambda_1 = 1, \lambda_2 = 3$$



$\lambda = 3$ has algebraic multiplicity of 2 in this case as this root appears twice in the characteristic polynomial.

Finding Eigenvectors

To find the eigenvectors, first compute the eigenvalues.

Then, for each eigenvalue, solve $(A - \lambda I)\underline{v} = \underline{0}$ for $\underline{v}$. These solutions form the eigenspace corresponding to this eigenvalue.

To get the eigenvectors, find a basis of this eigenspace

Example 1

$$A = \begin{pmatrix} 3 & 1 & 3 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

From previous section, $\lambda_1 = 1, \lambda_2 = 3$

For $\lambda_1 = 1$:

$$(A - \lambda I)\underline{v} = \underline{0}$$

$$(A - I)\underline{v} = \underline{0}$$

$$\begin{pmatrix} 2 & 1 & 3 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{pmatrix} \underline{v} = \underline{0}$$

$$\underline{v} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \text{ for some } x, y, z$$



For an eigenspace to be a vector space, it must contain the zero vector to provide an identity for vector addition, so the solution $\underline{v} = \underline{0}$ is actually included in the eigenspace, but its inclusion does not affect the computation of the basis (as $\underline{0}$ can be written as zero times any vector) so it can be ignored !!

Gauss-Jordan Time!

$$\left(\begin{array}{ccc|c} 2 & 1 & 3 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right) \rightarrow \left(\begin{array}{ccc|c} 1 & 0 & \frac{3}{2} & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

Steps: $\frac{1}{2} \times R_2$
 $R_1 - R_2$
 $\frac{1}{2} \times R_1$

$$\left. \begin{array}{l} x + \frac{3}{2}z = 0 \\ y = 0 \\ \text{let } z = \mu \in \mathbb{R} \\ x = -\frac{3}{2}\mu \end{array} \right\} \vec{z} = \begin{pmatrix} -\frac{3}{2}\mu \\ 0 \\ \mu \end{pmatrix} \rightarrow \text{but all vectors in the direction of this } \vec{z} \text{ are in the eigenspace, so the eigenspace contains the span of } \vec{z}.$$

→ Note any vector of this form is an eigenvector, can miss out the next step, but it can come in handy when geometric multiplicity > 1 to ensure linear independence of chosen eigenvectors.

∴ Eigenspace = $\text{Span}\left(\left\{\mu \begin{pmatrix} -\frac{3}{2} \\ 0 \\ 1 \end{pmatrix}\right\}\right) \rightarrow$ This eigenspace has dimension of 1. This means the eigenvalue $\lambda=1$ has geometric multiplicity of 1.

$$= \text{Span}\left(\left\{\begin{pmatrix} -\frac{3}{2} \\ 0 \\ 1 \end{pmatrix}\right\}\right)$$

Could use any vector in the span here, as any one vector qualifies as a basis for this eigenspace.

∴ eigenvector corresponding to $\lambda=1$ is $\begin{pmatrix} -\frac{3}{2} \\ 0 \\ 1 \end{pmatrix}$

Now to do that again with $\lambda=3$

For $\lambda=3$

$$(A - 3I)\underline{v} = \underline{0}$$

$$\left(\begin{array}{ccc|c} 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -2 & 0 \end{array}\right) \underline{v} = \underline{0} \quad \underline{v} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

Gauss-Jordan Time 2: The Sequel

$$\left(\begin{array}{ccc|c} 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -2 & 0 \end{array}\right) \rightarrow \left(\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

Steps: $-\frac{1}{2} \times R_3$
 $R_1 - 3 \times R_3$
 Swap $R_2 + R_3$

$$y = 0 \\ z = 0$$

Since only 0s in the first column, we can take any real value

$$x = \mu \in \mathbb{R} \rightarrow \underline{v} = \begin{pmatrix} \mu \\ 0 \\ 0 \end{pmatrix} = \mu \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\therefore \text{Eigenspace} = \text{Span}\left(\left\{\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}\right\}\right)$$

∴ eigenvector corresponding to $\lambda=3$ is $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \rightarrow \lambda=3$ has a geometric multiplicity of 1, even though its algebraic multiplicity is 2.

Note: Algebraic multiplicity of an eigenvalue is the number of times it appears as a solution to the characteristic polynomial
 Geometric multiplicity of an eigenvalue is the dimension of the eigenspace corresponding to that eigenvalue.

Example 2

$$\text{Let } A = \begin{pmatrix} 1 & 4 & 0 \\ 0 & 1 & 0 \\ 0 & 6 & 1 \end{pmatrix}$$

Compute eigenvalues:

$$\det(A - \lambda I) = \det \begin{pmatrix} 1-\lambda & 4 & 0 \\ 0 & 1-\lambda & 0 \\ 0 & 6 & 1-\lambda \end{pmatrix} = 0$$

$$(1-\lambda)(1-\lambda)(1-\lambda) = 0$$

$$\lambda = 1 \rightarrow \text{algebraic multiplicity } 3$$

Find eigenspace of eigenvalues:

$$\lambda = 1:$$

$$(A - \lambda I)\underline{v} = \underline{0}$$

$$(A - I)\underline{v} = \underline{0}$$

Gauss-Jordan

$$\left(\begin{array}{ccc|c} 0 & 4 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

Steps: $\frac{1}{4} \times R_1$

$\frac{1}{6} \times R_3$

$R_3 - R_1$

$$y = 0$$

Since only 0s in first and third column, both x and z are free variables, i.e. can take any real value, independent of each other.

$$\therefore x = \mu \in \mathbb{R} \rightarrow \underline{v} = \begin{pmatrix} \mu \\ 0 \\ \mu \end{pmatrix} = \mu \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$z = \phi \in \mathbb{R}$$

(logic flashbacks)

$$\text{Eigenspace} = \text{Span} \left(\mu \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \phi \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right) = \text{Span} \left(\left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\} \right)$$

This is equivalent to

Saying any linear combination

of $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$, i.e. the

Span of these 2 vectors

Find eigenvectors:

Since eigenvectors are the basis vectors for the eigenspace, $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ are eigenvectors corresponding to $\lambda = 1$

$\hookrightarrow$ This means $\lambda = 1$ has geometric multiplicity of 2.

Note: The geometric multiplicity of an eigenvalue is always less than or equal to its algebraic multiplicity.

Complex Eigenstuff

Eigenvalues and eigenvectors can also be complex numbers, written in the form $a + bi$, where $a, b \in \mathbb{R}$ and i is the imaginary number that some bored mathematician came up with to torture anyone learning maths, where $i^2 = -1$. As scary as they sound, they actually aren't that hard - just treat i like any other variable and substituting $i^2 = -1$ whenever relevant.

Example:

$$\text{find eigenvalues/vectors of } A = \begin{pmatrix} 3 & 0 & 0 \\ 2 & 1 & -1 \\ 0 & 2 & 3 \end{pmatrix}$$

Eigenvalues

Characteristic polynomial

$$\det(A - \lambda I) = 0$$

$$\det \begin{pmatrix} 3-\lambda & 0 & 0 \\ 2 & 1-\lambda & -1 \\ 0 & 2 & 3-\lambda \end{pmatrix} = 0$$

$$(3-\lambda)((1-\lambda)(3-\lambda)+2) = 0$$

$$(3-\lambda)(\lambda^2 - 4\lambda + 5) = 0$$

$$\lambda_1 = 3 \quad \lambda = \frac{4 \pm \sqrt{16-20}}{2}$$

$$\lambda = \frac{4 \pm \sqrt{-4}}{2} = \frac{4 \pm \sqrt{4}\sqrt{-1}}{2} = \frac{4 \pm 2i}{2}$$

$$= 2 \pm i$$

$$\sqrt{-4} = 2i, \sqrt{-9} = 3i \text{ etc.}$$

$$(2i)^2 = 4 \times (-1) = -4, \text{ etc.}$$

$\therefore$ Eigenvalues $\lambda_1 = 3, \lambda_2 = 2+i, \lambda_3 = 2-i$

for $\lambda_1 = 3$

$$(A - \lambda I)\underline{v} = \underline{0} \quad (\underline{v} = \begin{pmatrix} x \\ y \\ z \end{pmatrix})$$

$$\begin{pmatrix} 0 & 0 & 0 & | & 0 \\ 2 & -2 & -1 & | & 0 \\ 0 & 2 & 0 & | & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 2 & 0 & -1 & | & 0 \\ 0 & 1 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & -1/2 & | & 0 \\ 0 & 1 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{pmatrix}$$

Steps: $R_2 + R_3$

$1/2 \times R_3$

Swap R_2 and R_1

Swap R_3 and R_1

$1/2 \times R_1$

$$x - 1/2 z = 0 \rightarrow \text{let } z = \mu \in \mathbb{R}$$

$$y = 0$$

$$\therefore \text{eigenspace } \text{Span}\left(\left\{\begin{pmatrix} 1/2 \\ 0 \\ 1 \end{pmatrix}\right\}\right)$$

$$\therefore \text{eigenvector corresponding to } \lambda = 3: \begin{pmatrix} 1/2 \\ 0 \\ 1 \end{pmatrix}$$

for $\lambda_2 = 2+i$

Same as usual:

$$(A - \lambda I)\underline{v} = \underline{0}$$

$$\begin{pmatrix} 3-(2+i) & 0 & 0 \\ 2 & 1-(2+i) & -1 \\ 0 & 2 & 3-(2+i) \end{pmatrix} \underline{v} = \underline{0}$$

$$\begin{pmatrix} 1-i & 0 & 0 \\ 2 & -1-i & -1 \\ 0 & 2 & 1-i \end{pmatrix} \underline{v} = \underline{0}$$

Gauss-Jordan

$$\begin{pmatrix} 1-i & 0 & 0 & | & 0 \\ 2 & -1-i & -1 & | & 0 \\ 0 & 2 & 1-i & | & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 0 & | & 0 \\ 0 & 1+i & 1 & | & 0 \\ 0 & 2 & 1-i & | & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 0 & | & 0 \\ 0 & 1 & \frac{1-i}{2} & | & 0 \\ 0 & 2 & 1-i & | & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 0 & | & 0 \\ 0 & 1 & \frac{1-i}{2} & | & 0 \\ 0 & 0 & 0 & | & 0 \end{pmatrix}$$

$$\frac{1}{1-i} \times R_1$$

$$R_2 - 2R_1$$

$$-1 \times R_2$$

$$\frac{1}{1+i} \times R_2$$

$$\text{Note: } \frac{1}{1+i} = \frac{1-i}{(1+i)(1-i)} = \frac{1-i}{2}$$

$$(a+bi)(a-bi) = a^2 + b^2 \rightarrow \text{useful in removing } i \text{ from denominator.}$$

$$x = 0$$

$$y + \frac{1-i}{2} z = 0 \rightarrow y = \frac{-(1-i)}{2} z$$

$$\text{let } z = \mu \in \mathbb{R}$$

$$y = \left(\frac{-1+i}{2}\right)\mu$$

$$\text{Eigenspace} = \text{Span} \left(\left\{ \begin{pmatrix} 0 \\ -1+i \\ 1 \end{pmatrix} \right\} \right)$$

$\therefore$ eigenvector corresponding to $\lambda = \begin{pmatrix} 0 \\ -1+i \\ 2 \end{pmatrix}$ $\rightarrow$ can use any scalar multiple as before so $\begin{pmatrix} 0 \\ -1+i \\ 2 \end{pmatrix}$ would've looked less ugly but it doesn't matter. (looks don't matter :))

For $\lambda_3 = 2-i$:

$$(A - \lambda I)\underline{v} = \underline{0}$$

$$\begin{pmatrix} 3-(2-i) & 0 & 0 \\ 2 & 1-(2-i) & -1 \\ 0 & 2 & 3-(2-i) \end{pmatrix} \underline{v} = \underline{0}$$

$$\begin{pmatrix} 1+i & 0 & 0 \\ 2 & -1+i & -1 \\ 0 & 2 & 1+i \end{pmatrix} \underline{v} = \underline{0}$$

Gauss-Jordan

$$\begin{pmatrix} 1+i & 0 & 0 & | & 0 \\ 2 & -1+i & -1 & | & 0 \\ 0 & 2 & 1+i & | & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 0 & | & 0 \\ 0 & 1-i & 1 & | & 0 \\ 0 & 2 & 1+i & | & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 0 & | & 0 \\ 0 & 1 & \frac{1+i}{2} & | & 0 \\ 0 & 2 & 1+i & | & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 0 & | & 0 \\ 0 & 1 & \frac{1+i}{2} & | & 0 \\ 0 & 0 & 0 & | & 0 \end{pmatrix}$$

$\frac{1}{1+i} \times R_1$ $\frac{1}{1-i} \times R_2$ $R_3 - 2 \times R_2$

$$\begin{aligned} x &= 0 \\ y + \frac{1+i}{2}z &= 0 \Rightarrow y = -\frac{1+i}{2}z \\ \text{let } z &= \mu \in \mathbb{R} \\ y &= -\frac{1+i}{2}\mu \end{aligned}$$

$$\text{Eigenspace} : \text{Span} \left(\left\{ \begin{pmatrix} 0 \\ -1-i \\ 2 \end{pmatrix} \right\} \right)$$

$$\text{Eigenvector corresponding to } \lambda_3 = 2-i : \begin{pmatrix} 0 \\ -1-i \\ 2 \end{pmatrix}$$

Shortcut: if a matrix A with all real entries has a complex eigenvalue λ with corresponding eigenvectors $\underline{x}_1, \underline{x}_2, \dots$, then $\bar{\lambda}$ is also an eigenvalue of A with corresponding eigenvectors $\bar{\underline{x}}_1, \bar{\underline{x}}_2, \dots$. This saves unnecessary workings (as above) when working with complex eigenvalues $\rightarrow$ instead of manually computing the eigenvectors of λ_3 in the above example, can just state them as they are the conjugates of the eigenvectors of λ_2 .

Note: $\bar{\lambda}$ is the complex conjugate of λ , which is the same as λ but the sign of the imaginary part is flipped
(i.e. if $\lambda = a+bi$, $\bar{\lambda} = a-bi$; if $\underline{x} = \begin{pmatrix} a+bi \\ c+di \end{pmatrix}$, $\bar{\underline{x}} = \begin{pmatrix} a-bi \\ c-di \end{pmatrix}$)

$\downarrow$
This is an official Karl Swenson Certified method - he just stepped over it while speedrunning through the content.

Matrix Diagonalization

A matrix A is diagonalizable if there exists an invertible matrix P such that $P^{-1}AP$ is a diagonal matrix, i.e. only entries on the diagonal are non-zero, e.g. $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$. This is equivalent to saying A is diagonalizable if $A = PDP^{-1}$, where P is invertible and D diagonal.

Note: An $n \times n$ matrix is diagonalizable if and only if it has n linearly independent eigenvectors.

Method of diagonalizing an $n \times n$ matrix A :

- ① Find eigenvalues of A
- ② Find a basis for each eigenspace and put these basis vectors into a set S → i.e. find all linearly independent eigenvectors for each space and add them to S . So cannot use both $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$ for example
- ③ If S has fewer than n vectors, A is not diagonalisable.
Else, form the matrix P , with columns corresponding to the vectors in S
- ④ Calculate $D = P^{-1}AP$, which is diagonal

If A diagonalisable, S will contain n linearly independent vectors $\{x_1, x_2, \dots, x_n\}$. → this can be proven but who cares?

As a result $P = \begin{pmatrix} x_1 & x_2 & \dots & x_n \\ 1 & 1 & \dots & 1 \end{pmatrix}$ is invertible

Example:

Diagonalise $A = \begin{pmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix}$

① Find eigenvalues:

Char. eqn: $\det(A - \lambda I) = 0$

$$\det \begin{pmatrix} -\lambda & 0 & -2 \\ 1 & 2-\lambda & 1 \\ 1 & 0 & 3-\lambda \end{pmatrix} = 0$$

$$-\lambda(2-\lambda)(3-\lambda) + 2(2-\lambda) = 0$$

$$-\lambda(\lambda^2 - 5\lambda + 6) + 4 - 2\lambda = 0$$

$$-\lambda^3 + 5\lambda^2 - 6\lambda - 2\lambda + 4 = 0$$

$$\lambda^3 - 5\lambda^2 + 8\lambda - 4 = 0$$

$$(\lambda-1)(\lambda-2)^2 = 0$$

$$\lambda_1 = 1, \lambda_2 = 2$$

② Find bases of eigenspaces, i.e. distinct eigenvectors.

for $\lambda_1 = 1$

$$(A - I)x = 0$$

$$\begin{pmatrix} -1 & 0 & -2 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix} x = 0$$

$$G \sim \left(\begin{array}{ccc|c} -1 & 0 & -2 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 2 & 0 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

$$\begin{array}{l} \xrightarrow{R_3 + R_1} \\ -1 \times R_1 \end{array}$$

$$\begin{array}{l} x + 2z = 0 \\ y - z = 0 \end{array}$$

$$\text{let } z = \mu \in \mathbb{R}$$

$$y = \mu$$

$$x = -2\mu$$

$$\text{eigenspace: } \text{Span} \left\{ \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} \right\}$$

$$\text{basis } \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}$$

for $\lambda_2 = 2$

$$(A - 2I)v = 0$$

$$\begin{pmatrix} -2 & 0 & -2 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix} v = 0$$

$$\xrightarrow{G-J} \begin{pmatrix} -2 & 0 & -2 & | & 0 \\ 1 & 0 & 1 & | & 0 \\ 1 & 0 & 1 & | & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & 1 & | & 0 \\ 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{pmatrix}$$

$\rightarrow$

$-1/2 \times R_1$

$R_2 - R_1$

$R_3 - R_1$

$$x + z = 0$$

$$y = \mu \in \mathbb{R}$$

$$x + z = \phi \in \mathbb{R}$$

$$x = -\phi$$

$\rightarrow$ Since two independent free variables, need to split into two vectors to find the basis.

$$\text{Eigenspace} = \text{Span} \left\{ \begin{pmatrix} -\phi \\ \mu \\ \phi \end{pmatrix} \right\} = \text{Span} \left\{ \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}$$

$$= \begin{pmatrix} -\phi \\ 0 \\ \phi \end{pmatrix} + \begin{pmatrix} 0 \\ \mu \\ 0 \end{pmatrix} = \phi \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\Rightarrow \text{Spanning set } \left\{ \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}$$

$$\text{basis } \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$S = \left\{ \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\} \rightarrow 3 \text{ vectors for a } 3 \times 3 \text{ matrix}$$

$\therefore$ diagonalizable

$$\textcircled{3} P = \begin{pmatrix} -2 & -1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

$$\textcircled{4} P^{-1} = \begin{pmatrix} -1 & 0 & -1 \\ 1 & 0 & 2 \\ 1 & 1 & 1 \end{pmatrix}$$

Can use G-J to find P^{-1}
(or be lazy like me and use a calculator)

$$D = P^{-1}AP = \begin{pmatrix} -1 & 0 & -1 \\ 1 & 0 & 2 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix} \begin{pmatrix} -2 & -1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

$\rightarrow$ Note: matrix multiplication is associative $\Downarrow$
So $(P^{-1}A)P = P^{-1}(AP)$

$$D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

Note: The non-zero entries in D are the eigenvalues corresponding to the eigenvector forming their respective column in P . This is always the case.

$$D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} \quad P = \begin{pmatrix} -2 & -1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

$\lambda = 1$ $\lambda = 2$ (from above)

Eigen decomposition

An $n \times n$ matrix A has an eigendecomposition if and only if A is diagonalizable.

It can be found by diagonalising A to find D , then rewriting A as PDP^{-1}

E.g. The eigendecomposition of $A = \begin{pmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix}$ from the previous example is:

$$A = \begin{pmatrix} -2 & -1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} -1 & 0 & -1 \\ 1 & 0 & 2 \\ 1 & 1 & 1 \end{pmatrix}$$